Contact

danielhardestylewis@gmail.com

www.linkedin.com/in/dhardestylewis (LinkedIn)
dlewis.ai (Portfolio)
github.com/dhardestylewis (Portfolio)
homecastr.com (Portfolio)

Top Skills

Database Administration

Data Analytics

RDBMS

Languages

Spanish (Professional Working)

French (Limited Working)

English (Native or Bilingual)

Honors-Awards

Fellowship

Publications

Artificial Intelligence for Modeling Complex Systems

Daniel Hardesty Lewis

TOP500 Supercomputing | Bagnold Medal research contributor | NSF 10 Big Ideas

New York City Metropolitan Area

Summary

I have deep experience applying high-performance computing techniques to terrain and flood models, developing reproducible and scalable workflows from source data all the way out to web services, at scales as small as a parcel and as large as countries. I am currently developing the next generation of highest quality terrain data. I have a broad base of experience in research engineering, front- and back-end web development, and data analysis. Some of the reproducible water models that I have developed have more than 50,000 downloads from the hydro community. I am also highly active in my local community, serving as a catalyst for positive change on behalf of and impacting more than 10,000 residents.

Experience

South Park Commons
Member

June 2026 - Present (1 month)

New York, NY

homecastr
Founder

September 2025 - Present (10 months)

New York, NY

forecasting property values 5 years out

homecastr.com

Summit Geospatial
Founder

November 2023 - August 2025 (1 year 10 months)

the highest quality elevation data in Texas

summitgeospatial.com

Texas Advanced Computing Center (TACC)

5 years 7 months

Senior Data Scientist and Technical Lead

August 2021 - August 2023 (2 years 1 month)

Austin, Texas, United States

Millions of dollars in compute

Million-node jobs

Millions of earth images produced

All at the highest quality

Contributed research towards Paola Passalacqua's 2022 Bagnold Medal, one of the highest honors in geomorphology
https://www.egu.eu/awards-medals/ralph-alger-bagnold/2022/paola-passalacqua/

Led a principal project of $40M disaster resiliency initiative, TDIS
https://www.glo.texas.gov/disaster-recovery/planning-studies/texas-disaster-information-system

Contributed to an NSF Big Idea
https://www.nsf.gov/about/history/big-ideas#navigating-the-new-arctic-3f4

I design and implement intelligent decision support systems to support computational science, disaster mitigation and response, and resiliency research projects.

- scale climate models on the most powerful computers in the world
- develop relationships with university, state, and federal agencies (NOAA, US Army Corps of Engineers, Texas General Land Office, Texas General Information Office, Texas Water Development Board, TNRIS, USC, UT, A&M, etc)
- prepare computing facility for rapid response to drought, hurricane, flood disasters with replicable scaling of engineering workflows and training

Data Scientist and Research Engineer

February 2018 - July 2021 (3 years 6 months)

Austin, Texas, United States

Deep Learning for Petrobras

DARPA World Modelers: disaster resiliency in East Africa https://www.darpa.mil/research/programs/world-modelers

I contribute research towards integrated modelling and water resource management within the state of Texas and abroad, and I've applied machine and deep learning approaches to problems for the energy industry.

The University of Texas at Austin
Co-instructor
January 2018 - December 2020 (3 years)
Austin, Texas, United States

Machine Learning for the Geosciences
Intelligent Systems for the Geosciences
Scientific Computation

ICC Austin
6 years

System Architect
September 2014 - August 2020 (6 years)
Austin, Texas, United States

Re-designed and implemented entirely new internet infrastructures for 9 buildings serving 115 students, enabling gigabit speeds at each. Re-positioned new enterprise-level wireless access points. Trained organizational capacity to manage new infrastructure.

Conflict mediator
June 2015 - August 2018 (3 years 3 months)
Austin, Texas

Mediated disputes between members and their housing coops. Familiar with all applicable policies and bylaws. Guided facilitations which reached mutually agreed-upon solutions, to help ensure the continued success of this tenant self-managed housing organization.

Technology Officer
September 2014 - December 2017 (3 years 4 months)
Austin, Texas area

Set up and managed public computers for a 20+ tenant building.

ICPSR at University of Michigan Institute for Social Research
Program Scholar
June 2019 - July 2019 (2 months)
Ann Arbor, Michigan, United States

The ICPSR Summer Program provides rigorous, hands-on training in statistical techniques, research methodologies, and data analysis. ICPSR Summer Program courses emphasize the integration of methodological strategies with the theoretical and practical concerns that arise in research on substantive issues.

Courses enrolled:
- Machine Learning: Applications and Opportunities in the Social Sciences
- Intro to Network Analysis
- Math for Social Science III

Texas Advanced Computing Center (TACC)
Undergraduate Research Assistant
May 2017 - August 2017 (4 months)
Austin, Texas, United States

With a $4000 fellowship from the Texas Institute for Discovery Education in Science, I helped update Texas's groundwater availability models to a more contemporary format

MICEFA
Information Technology Intern
October 2016 - January 2017 (4 months)
Paris, Île-de-France, France

Developed Bash workflow to transform data from a proprietary database into SQL

The University of Texas at Austin
Undergraduate Research Assistant
May 2013 - December 2013 (8 months)
Austin, Texas Area

The boundaries of the stellar evolution code MESA were tested, especially with regard to the high-mass, oxygen/neon-core white dwarf regime. Theoretical models were run, and the existence (by the standard physics as implemented in the program) of such a regime was found from these calculations.

Education

Columbia University
Master of Science - MS · (2024 - 2026)

Columbia Engineering
Cross-registered · (2024 - 2026)

The University of Texas at Austin
Bachelor of Science - BS, Pure Mathematics · (2012 - 2017)

The University of Texas at Austin
Certificate, Scientific Computation